

Aleksei Nizhgorodov

Frontend Engineer

Moscow, Russia · open to relocation & remote

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SUMMARY

Four years in commercial development. I go for the hard parts: WebSocket real-time, data-heavy dashboards, maps, WebGL. When no library does the job I write my own; the latest prints WebGL maps into PDF reports straight from React components. I've led small frontend teams, and on some products I was the only frontend engineer. English B2. Open to new opportunities.

SKILLS

Core: TypeScript, JavaScript (ES6+), HTML, CSS

Frameworks: React, Svelte, Next.js, Node.js, Express

Maps & data-viz: Leaflet, WebGL, Canvas, ECharts, Recharts, QGIS

State & data: MobX, Redux, Zod, REST, WebSocket, SSE, PostgreSQL, Drizzle ORM

Build & infra: Vite, Webpack, Capacitor, PWA, Service Worker, Docker, Nginx, CI/CD (GitLab)

Testing: Vitest, React Testing Library, Playwright

EXPERIENCE

Perfema — Frontend Developer

Apr 2023 — Present · Moscow

Full-time job. I build products here from scratch.

- Team lead: ran frontend teams of 2–4, did code review, mentored and onboarded newcomers. Held technical meetings and demos with clients.
- Occupational-health analytics (B2B): built the MVP in about two weeks, then grew it to ~20 clinical dashboards. A pnpm monorepo: React with MobX, an Express BFF, shared Zod contracts. Ingest survives upstream outages and duplicate deliveries (backoff, circuit breaker), and read-layer RBAC keeps medical data out of the browser.
- Mapping & real-time monitoring: wrote a rendering layer on top of Leaflet with Canvas and WebGL. It holds thousands of live markers and long GPS tracks without dropping frames; clustering, geofences and a history timeline are built in. Several company products run on it now.
- Video-analytics platform: designed the frontend from scratch (TypeScript, React, MobX) and set the patterns for the team. Live dashboards on Recharts, virtualized tables for large datasets.
- PDF reports: wrote our own library after jsPDF and react-pdf choked on WebGL maps. It prints maps and charts from plain React components and handles page numbers and running headers by itself.
- Kids' task tracker: two clients, a big-screen dashboard and a PWA for parents, sharing state in real time. The team packaged it with Capacitor into native apps for iOS, Android and Android TV.
- Odds and ends: i18n with RTL Arabic, CryptoPro document signing in the browser, offline and auto-update via a service worker, an SSR site on Svelte and Strapi, GitLab CI/CD. Tests in Vitest, React Testing Library and Playwright.

Edya — Frontend Engineer (part-time)

2025 — Present · Remote

- VPN customer portal: designed and shipped the production SPA on React 19, TanStack Router and MobX, now at about 20k MAU. White-label: eight brands ship from one codebase, adding a brand is a config change. Cut the initial bundle from 1706 to 656 KB, gzip from 467 to 175.
- Censorship circumvention: on every navigation a service worker revalidates the mirror list and moves the user off a dead domain to a live one, keeping the path and query. A static gate page on GCS picks up visits from bot links and bookmarks once a domain is blocked. Both the app bundle and the gate are obfuscated.
- Also in the portal: sign-in via Google, email or Telegram, subscription transfer from the legacy Telegram bot, referrals, guest key activation with per-platform setup guides up to Android TV.
- Metrics: product events, exceptions and API failures go to PostHog, sliced by brand, platform and release.
- EdyaAI, a Telegram super-app: a Mini App with ChatGPT, Grok and DeepSeek chat plus Midjourney image generation in one place. I owned the generation engine, payments and the legacy-chat integration. Payments are live: plans, subscriptions, promo codes. The legacy Next.js part is embedded in an iframe with a full event bridge, haptics included, so users never notice the seam.
- Generation engine: WebSocket-based, a state machine per request, automatic reconnects. Live generation progress, upscales and variations, image references, merging two pictures, rate limits with a countdown to reset.

Links: edya.org

Yandex — Assessor-Developer

Jan — Dec 2023 · Moscow

- Labeled programming and CS material for YandexGPT: wrote and edited reference answers on IT topics, checked facts.

EDUCATION

HSE University — MSc, Digital Urbanism & City Analytics

2024 — 2026 · Moscow

Data analytics, Python, geodata, QGIS. Thesis: hse.ru/edu/vkr/1167123484

NUST MISIS — BSc, Information Systems & Technologies

2020 — 2024 · Moscow

Algorithms, data structures, software engineering. Hackathon prizes: 2nd place at the LCT 2023 hackathon (misis.ru/news/8829), MTS True Tech Hack, Tsifra Fest, X-MAS Hack.

LANGUAGES

Russian — Native · English — B2

FUN FACT

Claude or Codex? Claude. Personal records: 40M tokens in a month and 6M in a single request.